

Tide Display - Manual

This is a tidal display, relying on mathematical models for all of its predictions, allowing the device to operate offline. Not needing wireless connections and employing a low refreshrate e-paper display, the device is capable of running for 2+ years with AA batteries.

1. Moon Phase Display

The current moon phase is calculated via the Meeus algorithm, which uses the 29.53-day lunar cycle and applies 15 periodic corrections for the gravitational pull of the sun, the moon's elliptical orbit, and other astronomical factors. It is accurate to within 1% indefinitely.

2. Next Full and New Moon Dates

Using the same mathematical model, the next full and new moon dates are calculated. The predictions are accurate to within 2 minutes indefinitely.

3. Sunrise and Sunset Times

Based on another equation derived by Jean Meeus, the model takes into account the Earth's elliptical orbit and axial tilt to determine sunrise and sunset times for a given latitude. The algorithm predicts within an error of 2 minutes. Local phenomena such as geographical features or tall buildings are not taken into account.

4. Daily Daylight Display

Using the sunrise and sunset times, the bar represents a 24-hour cycle starting at midnight. Dark sections indicate

night; light sections indicate daylight.

5. Tidal Display

Unlike conventional tidal clocks that simply show a 12-hour 25-minute cycle, this tidal display is based on a harmonic model taking into account 31 periodic phenomena. The most influential are the orbits of the sun and moon; other factors include local effects such as shallow water and the Thames estuary. As the mathematical models needed for tide predictions are not freely available, a model was derived using historic tidal data and predictions from the Port of London Authority. The model achieves 10–15 minute accuracy for most days. Predictions will likely remain accurate over the next 25 years; however, changes in sea level and long-period astronomical effects may reduce accuracy in the distant future. Weather effects are not taken into account.

6. Daily Message

Shows important events including summer and winter solstices, equinoxes, solar and lunar eclipses visible from Margate over the next 100 years, and the six most prominent recurring meteor showers.

7. Location

Shows the location for which this display is calibrated. The sunrise, sunset, and tidal calculations are specific to this geographical location.

1. Moon phase display

Illuminated surface shown as white



○ 03/01
● 18/01

2. Next Full and New Moon dates

Full moon shown as white circle

07:58 ☀ 15:49

3. Sunrise and Sunset times

Maximum deviation of two minutes

15:15

4. Daily Daylight display

The bar represents 24 hours, white sections represent hours of sunlight



5. Tidal display

Shows the current tide, next high and low tide, flooding and ebbing.

21:10

6. Daily Message

Indicates events of the day

Merry Christmas!

7. Location

Location to which the display is calibrated to

Margate